

Mathematics for Social Scientists

Part 3: Linear Algebra I: Vectors and Matrices

Carlos Gueiros

University of Mannheim

Fall 2025

Outline

Motivation

Vectors

Vectors Operations

Matrices

Matrix Operations

Vector Spaces

Motivation

When one reads high visibility journals in the social sciences, matrix algebra is ubiquitous.

- Jeff Gill, *Essential Mathematics for Political and Social Research*

Motivation

Example (Powers and Cox (1997)).

A **system of equations** for relating political blame to various demographic and regional political variables:

for $i = 1$ to n ,

$$\begin{aligned} (BLAMEFIRST) Y_i = & \beta_0 + \beta_1 CHANGELIV + \beta_2 BLAMECOMM + \beta_3 INCOME \\ & + \beta_4 FARMER + \beta_5 OWNER + \beta_6 BLUESTATE \\ & + \beta_7 WHITESTATE + \beta_8 FORMMCOMM + \beta_9 AGE \\ & + \beta_{10} SQAGE + \beta_{11} SEX + \beta_{12} SIZEPLACE \\ & + \beta_{13} EDUC + \beta_{14} FINHS + \beta_{15} ED * HS \\ & + \beta_{16} RELIG + \beta_{17} NATION + E_i \end{aligned}$$

In matrix algebra form this is $Y = X\beta + \varepsilon$.

Motivation

- A statistical model describes how some variables ($x_0 \dots x_k$) generate another variable y given some parameters ($\beta_0 \dots \beta_k$) and an error term ($\epsilon_1 \dots \epsilon_n$), e.g. the linear regression model
- to estimate the parameters, we essentially set up a system of n equations
- each equation describes how each of our n data point was generated, e.g.

$$y_1 = \beta_0 + \beta_1 x_1 + \epsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_2 + \epsilon_2$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 x_n + \epsilon_n$$

Motivation: From many equations to one line

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}_Y = \underbrace{\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}}_X \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}}_{\beta} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{\varepsilon} \Rightarrow Y = X\beta + \varepsilon$$

- One line captures all n relationships: this is why matrix notation is so powerful.

Vectors

Definition 1.

- A **vector** is a list of real numbers where *order matters*, denoted by

$$\text{a column vector } v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \quad \text{or a row vector } v = (v_1, \dots, v_n),$$

where $v_1, \dots, v_n \in \mathbb{R}$.

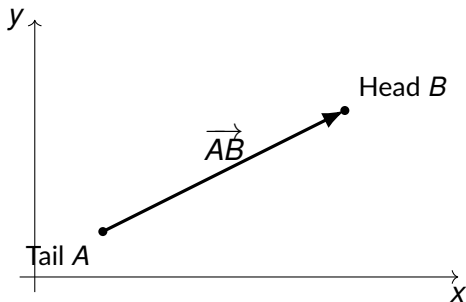
- The set containing all vectors of length n is called a **vector space**, and is denoted by $\mathbb{R}^n$.

Vectors: geometric interpretation

In analytic geometry, a vector can be viewed as a **directed line segment**:

- **Tail** (starting point) and **head** (ending point).
- **Magnitude** (length of the arrow).
- **Direction** (the way the arrow points).

We may write the same vector as an arrow $\overrightarrow{AB}$ or as a coordinate vector $\mathbf{v}$.

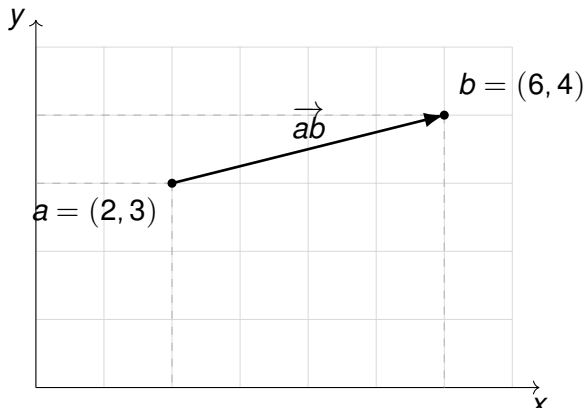


Vectors

A vector in the plane ($\mathbb{R}^2$) can be interpreted as the **vector from a to b** .

Let $a = (2, 3)$ and $b = (6, 4)$. The vector from a to b is

$$\vec{ab} = b - a = (b_1 - a_1, b_2 - a_2) = (4, 1) = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$



Vectors

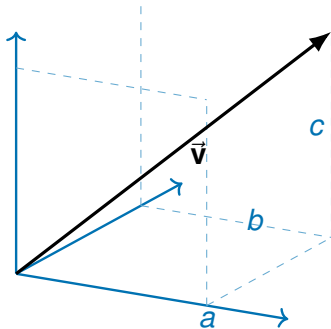


Figure: Representation of vectors in $\mathbb{R}^3$.

Consider the vector $\vec{v} = (a, b, c)$, or in column-matrix form

$$\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \quad a, b, c \in \mathbb{R}.$$

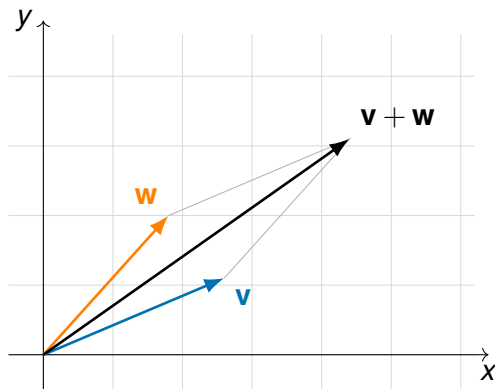
Geometric interpretation. $\vec{v}$ is the arrow from the origin $(0, 0, 0)$ to the point (a, b, c) .

Vector Addition

- **Vector addition:** $:\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$,
 $(v, w) \mapsto v + w$,

$$v + w = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{pmatrix}$$

Both vectors must be the same size (dimension).

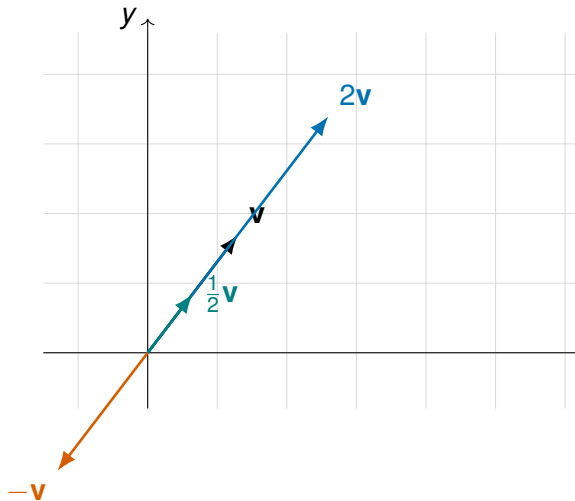


Scalar Multiplication

- **Vector scalar multiplication:**

$$\mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n, (\lambda, v) \mapsto \lambda \cdot v,$$

$$\lambda \cdot v = \lambda \cdot \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} \lambda \cdot v_1 \\ \vdots \\ \lambda \cdot v_n \end{pmatrix}$$



Examples

Example.

1.

$$\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$$

2.

$$3 \times \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

Some consequences:

- $0 \cdot v = \mathbf{0}$
- $\lambda \cdot \mathbf{0} = \mathbf{0}$
- $\lambda \cdot v = \mathbf{0} \Rightarrow \lambda = 0 \text{ or } v = \mathbf{0}$
- $(-1) \cdot v = -v$

Vector Norm and Distance

The **norm** (length) of a vector $\mathbf{a} = (a_1, \dots, a_n)$ is:

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2} = \sqrt{\sum_{i=1}^n a_i^2}.$$

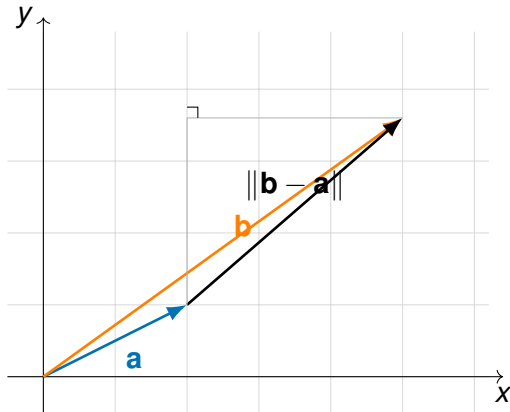
A **normalized vector** has norm 1. A **zero vector** has norm 0 (note: $\|\mathbf{a}\| = 0 \iff a_i = 0$ for all i).

Application in $\mathbb{R}^2$ (Euclidean distance between $\mathbf{a}$ and $\mathbf{b}$):

$$\|\mathbf{a} - \mathbf{b}\| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2} \quad (\text{Pythagoras}).$$

Generalized to n dimensions:

$$\|\mathbf{a} - \mathbf{b}\| = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}.$$

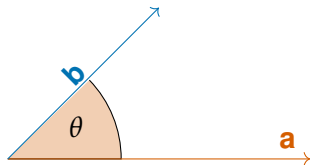


Dot product

The **inner product** (dot product) of two vectors of equal dimension is defined as:

$$\mathbf{a} \cdot \mathbf{b} = a_1 \cdot b_1 + a_2 \cdot b_2 \dots a_n \cdot b_n = \sum_{i=1}^n a_i b_i$$

Graphically ($\mathbb{R}^2$):



$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$, where θ is the **angle** between the vectors.

Properties

Properties of the Dot Product

If $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are n -vectors and α is a scalar, then

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$.
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$.
- $(\alpha \mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (\alpha \mathbf{b}) = \alpha(\mathbf{a} \cdot \mathbf{b})$.
- $\mathbf{a} \cdot \mathbf{a} > 0 \iff \mathbf{a} \neq \mathbf{0}$.

Matrices

Definition 2.

A **matrix** A with n rows and m columns is a rectangular array of numbers of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix}.$$

We call n and m the **dimensions** of the matrix, denoted $n \times m$, and the numbers a_{ij} are called the **entries**. If $n = m$, the matrix is called **square**.

Example: $\begin{pmatrix} 1 \\ 6 \\ -12 \end{pmatrix}$ is a 3×1 matrix and $\begin{pmatrix} 1 & 1 & 19 \\ 3 & 5 & -16 \end{pmatrix}$ is a 2×3 matrix.

Matrices

- Two matrices A and B are equal if and only if they have equal dimensions and every element of A is equal to the corresponding element of B :

$$a_{ij} = b_{ij} \quad \forall 1 \leq i \leq m, 1 \leq j \leq n.$$

- The set of all matrices with n rows and m columns is denoted by $\mathbb{R}^{n \times m}$.
- A column vector of length n is a matrix with dimensions $n \times 1$.

Matrix Addition

The addition of two matrices with the same dimensions is defined as follows:

$$: \mathbb{R}^{n \times m} \times \mathbb{R}^{n \times m} \rightarrow \mathbb{R}^{n \times m}, \quad (A, B) \mapsto A + B$$

where

$$\begin{aligned} A + B &= \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{pmatrix} \\ &= \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1m} + b_{1m} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2m} + b_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} + b_{n2} & a_{n2} + b_{n2} & \dots & a_{nm} + b_{nm} \end{pmatrix}. \end{aligned}$$

Multiplication of a matrix with a scalar

The multiplication of a matrix with $\lambda \in \mathbb{R}$ is defined as follows:

$$\mathbb{R} \times \mathbb{R}^{n \times m} \rightarrow \mathbb{R}^{n \times m}, \quad (\lambda, A) \mapsto \lambda \cdot A$$

where

$$\begin{aligned} \lambda \cdot A &= \lambda \cdot \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \\ &= \begin{pmatrix} \lambda \cdot a_{11} & \lambda \cdot a_{12} & \dots & \lambda \cdot a_{1m} \\ \lambda \cdot a_{21} & \lambda \cdot a_{22} & \dots & \lambda \cdot a_{2m} \\ \vdots & \vdots & & \vdots \\ \lambda \cdot a_{n1} & \lambda \cdot a_{n2} & \dots & \lambda \cdot a_{nm} \end{pmatrix}. \end{aligned}$$

Properties: Addition and Scalar Multiplication

1. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$

2. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$

3. $\mathbf{A} + \mathbf{0} = \mathbf{A}$

4. $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$

5. $(\lambda + \beta)\mathbf{A} = \lambda\mathbf{A} + \beta\mathbf{A}$

6. $\lambda(\mathbf{A} + \mathbf{B}) = \lambda\mathbf{A} + \lambda\mathbf{B}$

Examples

Example.

1.

$$\begin{pmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 3 & 1 \\ 0 & 1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ 2 & 1 \\ 3 & 2 \end{pmatrix}$$

2.

$$3 \times \begin{pmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 6 & 0 \\ 0 & 3 \end{pmatrix}$$

Q: What is the neutral element of $(\mathbb{R}^{n \times m}, +)$?

Special matrices

- The *zero matrix*:

$$0 = 0_{n \times m} = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

- *Diagonal matrices*: Square matrices with non-zero entries on the *diagonal* only:

$$A = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$$

Special matrices ctd.

- The *identity matrix*: The diagonal matrix with $a_{ij} = 1$ for $i = 1, \dots, n$.

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

- (Lower) *Triangular Matrix*: The matrix with $a_{ij} = 0$ for $i < j$.

$$\begin{pmatrix} a_{11} & a_{12} & \dots & \dots & a_{1m} \\ 0 & a_{22} & \dots & \dots & a_{2m} \\ 0 & 0 & a_{33} & \dots & a_{3m} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & a_{nm} \end{pmatrix}$$

Idempotent Matrix and Trace

Idempotent matrix. A square matrix A is **idempotent** if

$$A^2 = A.$$

Example:

$$\underbrace{\begin{pmatrix} 5 & -5 \\ 4 & -4 \end{pmatrix}}_A \times \underbrace{\begin{pmatrix} 5 & -5 \\ 4 & -4 \end{pmatrix}}_A = \begin{pmatrix} 5 & -5 \\ 4 & -4 \end{pmatrix} = A.$$

Trace. The **trace** of a square matrix is the sum of its main diagonal entries:

$$\text{tr}(A) = \sum_i a_{ii}.$$

Example:

$$\text{tr} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 1 + 5 + 9 = 15.$$

Correlation matrices

Matrices provide a convenient way to display correlations between many different variables. **Correlation matrices** are symmetrical along the diagonal, and so are often presented in triangular form.

Example (Norton, (1983)).

- a: We have a good marriage.
- b: My relationship with my partner is very stable.
- c: Our marriage is strong.
- d: My relationship with my partner makes me happy.
- e: I really feel like *part of a team* with my partner.
- f: The degree of happiness, everything considered. (10-point scale).

	a	b	c	d	e	f
a	1.00	0.85	0.83	0.83	0.74	0.76
b		1.00	0.82	0.86	0.72	0.77
c			1.00	0.78	0.68	0.70
d				1.00	0.71	0.76
e					1.00	0.69
f						1.00

Transpose of a matrix

Definition 3.

Let $A = (a_{ij})_{i=1,\dots,n,j=1,\dots,m} \in \mathbb{R}^{n \times m}$ be a matrix. Then, the matrix $A^\top \in \mathbb{R}^{m \times n}$ defined via

$$A^\top := (a_{ji})_{j=1,\dots,m,i=1,\dots,n} \in \mathbb{R}^{m \times n}$$

is called the **transpose of A** .

- ... that is, transpose is formed by switching rows and columns.
 - First row becomes first column, second row second column and so on ...

Transpose and Symmetry

Example.

Let

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{pmatrix}. \quad \text{Then} \quad B^{\top} = \begin{pmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{pmatrix}.$$

Properties of the transpose

1. $(A^{\top})^{\top} = A$
2. $(A + B)^{\top} = A^{\top} + B^{\top}$
3. $(\alpha A)^{\top} = \alpha A^{\top}$
4. $(AB)^{\top} = B^{\top} A^{\top}$

Definition 4.

A (square) matrix A is **symmetric** if $A = A^{\top}$.

Multiplying a matrix and a vector

Let $A \in \mathbb{R}^{n \times m}$ and $v \in \mathbb{R}^m$. Then the product $Av = A \cdot v$ is

$$Av = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_m \end{pmatrix} = \begin{pmatrix} a_{11}v_1 + a_{12}v_2 + \dots + a_{1m}v_m \\ a_{21}v_1 + a_{22}v_2 + \dots + a_{2m}v_m \\ \vdots \\ a_{n1}v_1 + a_{n2}v_2 + \dots + a_{nm}v_m \end{pmatrix}.$$

Multiplying a matrix and a vector ctd

- The product Av is only defined when the number of columns of A equals the number of rows of v . The result Av has the same number of rows as A and one column.

$$\underset{n \times m}{\text{matrix}} \times \underset{m}{\text{vector}} = \underset{n}{\text{vector}}$$

Example.

$$\begin{pmatrix} 4 & 2 & 3 \\ -1 & 6 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = ?$$

Some questions

Q: What matrix A fulfills the identity $Av = v$ for all $v \in \mathbb{R}^n$?

Q: What matrix A fulfills the identity $Av = \mathbf{0}$ for all $v \in \mathbb{R}^n$?

But notice e.g.

$$\begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$
$$\begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Multiplication of two matrices

For two matrices $A \in \mathbb{R}^{r \times n}$ and $B \in \mathbb{R}^{n \times s}$ the product AB is defined via

$$AB = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{r1} & a_{r2} & \dots & a_{rn} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1s} \\ b_{21} & b_{22} & \dots & b_{2s} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \dots & b_{ns} \end{pmatrix}$$
$$= \begin{pmatrix} a_{11}b_{11} + \dots + a_{1n}b_{n1} & a_{11}b_{12} + \dots + a_{1n}b_{n2} & \dots & a_{11}b_{1s} + \dots + a_{1n}b_{ns} \\ a_{21}b_{11} + \dots + a_{2n}b_{n1} & a_{21}b_{12} + \dots + a_{2n}b_{n2} & \dots & a_{21}b_{1s} + \dots + a_{2n}b_{ns} \\ \vdots & \vdots & & \vdots \\ a_{r1}b_{11} + \dots + a_{rn}b_{n1} & a_{r1}b_{12} + \dots + a_{rn}b_{n2} & \dots & a_{r1}b_{1s} + \dots + a_{rn}b_{ns} \end{pmatrix}$$

Multiplication of two matrices ctd.

- The product AB is defined only if the number of columns of the left matrix (here: A) is equal to the number of rows of the right matrix (here: B). The result has the same number of rows as A and the same number of columns as B .

$$\begin{matrix} \text{matrix} & \cdot & \text{matrix} & = & \text{matrix} \\ r \times n & & n \times s & & r \times s \end{matrix}$$

Example.

$$\begin{pmatrix} 1 & 0 & 3 & 0 \\ 2 & 1 & 2 & 1 \\ 5 & 2 & 0 & 1 \end{pmatrix}_{3 \times 4} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 4 & 0 & 3 \\ 2 & 2 & 1 \\ 0 & 3 & 1 \end{pmatrix}_{4 \times 3} = \begin{pmatrix} 7 & 7 & 3 \\ 10 & 9 & 6 \\ 13 & 8 & 7 \end{pmatrix}_{3 \times 3}$$

Example.

Order matters when multiplying matrices!

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 9 \\ 1 & 3 \end{pmatrix}$$
$$\begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 7 \\ 1 & 6 \end{pmatrix}$$

Some rules for matrix multiplication

Let $A, A' \in \mathbb{R}^{r \times n}$, $B, B' \in \mathbb{R}^{n \times s}$, $C \in \mathbb{R}^{s \times t}$, $\lambda \in \mathbb{R}$. Then:

- $A \cdot (B + B') = A \cdot B + A \cdot B'$ and $(A + A') \cdot B = A \cdot B + A' \cdot B$,
- $A \cdot (\lambda B) = (\lambda A) \cdot B = \lambda(A \cdot B)$,
- $(A \cdot B) \cdot C = A \cdot (B \cdot C)$,
- $(A \cdot B)^\top = B^\top \cdot A^\top$,
- $I_r \cdot A = A = A \cdot I_n$.

Vector Space

Definition 5.

Vector space. A (nonempty) set V with two operations—*vector addition* and *scalar multiplication by real numbers*—is a vector space if, for any vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in V$ and scalars $a, b \in \mathbb{R}$, the following hold:

- $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ (associativity of addition)
- $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ (commutativity)
- $\exists \mathbf{0} \in V$ s.t. $\mathbf{a} + \mathbf{0} = \mathbf{a}$ (additive identity)
- $\forall \mathbf{a} \exists (-\mathbf{a})$ s.t. $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$ (additive inverse)
- $a(\mathbf{b} + \mathbf{c}) = a\mathbf{b} + a\mathbf{c}$ (distributivity over vector addition)
- $(a + b)\mathbf{a} = a\mathbf{a} + b\mathbf{a}$ (distributivity over scalar addition)
- $a(b\mathbf{a}) = (ab)\mathbf{a}$ (associativity of scalar multiplication)
- $1\mathbf{a} = \mathbf{a}$ (multiplicative identity)

Examples of vector spaces (over $\mathbb{R}$)

- $\mathbb{R}^2 = \{(x_1, x_2) : x_i \in \mathbb{R}\}$
- $\mathbb{R}^3 = \{(x_1, x_2, x_3) : x_i \in \mathbb{R}\}$
- $\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$ (column or row vectors)
- $M_{m \times n} = \{m \times n \text{ real matrices}\}$
- $M_{2 \times 2} = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}$

Subspaces

Subspace. A subset $W \subseteq V$ (with V a vector space over $\mathbb{R}$) is a subspace if it is nonempty and

$$\mathbf{u}, \mathbf{v} \in W, a, b \in \mathbb{R} \implies a\mathbf{u} + b\mathbf{v} \in W.$$

Examples:

- $\{\mathbf{0}\}$ and V itself.
- Any line through the origin in $\mathbb{R}^2$; any plane through the origin in $\mathbb{R}^3$.

Non-example. $\{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 = 1\}$ (does not contain $\mathbf{0}$; not closed under scaling).

Linear combinations and span

Linear combination. Given $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$, any vector of the form

$$a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k \quad (a_1, \dots, a_k \in \mathbb{R})$$

is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_k$.

Span. The set of all linear combinations is

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k : a_i \in \mathbb{R}\},$$

which is a subspace and the *smallest* subspace of V that contains $\mathbf{v}_1, \dots, \mathbf{v}_k$.